

Analysis PhD Exam, August 2012

Be sure to carefully present all work in a neat and logical fashion. Do not leave any gaps. State clearly theorems used in your proofs.

1. This problem has two parts.
 - (a) State the Vitali convergence theorem (define all terms).
 - (b) Use this theorem to prove the Lebesgue Dominated Convergence theorem (first state this theorem).
2. Let f be an integrable function on (S, Σ, μ) . Let ν be the indefinite integral of f . Let $|\nu|(\cdot)$ denote the total variation of ν defined on Σ . Show that $|\nu|(\cdot)$ is the indefinite integral of an a.e. unique integrable function g . What is the relationship between g and f ? Prove.
3. Let (S, Σ, μ) be a finite measure space. Suppose x^* belongs to the dual space of $L^1(S, \Sigma, \mu)$. Show that there exist an a.e. unique integrable function g such that

$$x^*(f) = \int_S fg \, d\mu, \quad \text{for each } f \in L^1(S, \Sigma, \mu).$$

[You need not prove that $g \in L^\infty$, etc].

4. Let $\{y_n\}$ be an orthogonal sequence in a Hilbert space. Show $\sum y_n$ converges unconditionally if and only if $\sum |y_n|^2 < \infty$.
5. Let X be a Banach space with closed linear subspaces Y and Z . Suppose each x in X is the unique sum $y + z$ where $y \in Y$ and $z \in Z$. Show that there is a constant K such that $|y| \leq K|x|$ and $|z| \leq K|x|$ for each x in X with representation $x = y + z$. [Hint: Use the open mapping theorem or closed graph theorem for certain maps]
6. Let $\{x_n\}$ be a weakly convergent sequence in a normed space X . Show that the weak limit belongs to the norm closed span of x_n .